Anchor-Mediated Transport for Credible Image Restoration

Restricting Restoration Trajectories Through an Evidence-Sufficient Intermediate Manifold

A controlled study on MNIST denoising · November 2026

Abstract

Modern image restoration is increasingly cast as a probabilistic transport problem: a flow-matching ODE, a diffusion SDE, or a two-marginal Schrödinger Bridge (SB) learns a trajectory from a degraded observation y to a clean image x. We argue that the persistent failure mode of these models — **ungrounded hallucination**, i.e. the invention of high-frequency details that are not supported by y — is a structural consequence of leaving the *intermediate* trajectory unconstrained. Only the two endpoints are pinned, so the transport optimizer can take statistically convenient shortcuts through the high-frequency texture manifold learned from the training distribution.

We propose **Anchor-Mediated Transport (AMT)**: instead of one unconstrained transport from y to x, we insert a mandatory intermediate anchor manifold $z = \mathrm{LowPass}(y)$ that retains only the observation-supported, SNR-rich low-frequency content and explicitly discards the noise-dominated high frequencies. Every restoration trajectory is factorized into two coupled segments, $y \to z$ and $z \to x$, and the second segment is trained with a low-frequency consistency term that forbids the network from re-inventing the structure already observed in y. On MNIST additive Gaussian denoising ($\sigma=0.5$), AMT reduces the structured residual signal (the observable proxy for hallucination) by **41%** relative to an unconstrained direct black-box regressor, at the cost of only ~2.5 dB PSNR and ~1.7 points of downstream digit-classification accuracy.

1. Introduction

The task. We study image restoration as an inverse problem. Concretely, let x be an unknown clean image and y the degraded observation generated by an imaging operator D plus noise: $y = D(x) + n$. In this paper we instantiate the canonical denoising case D = identity, $n \sim \mathrm{N}(0, \sigma^2)$. Because the forward operator is many-to-one, the inverse problem is ill-posed: infinitely many clean images are consistent with a given y.

Why black-box regression hallucinates. A conventional feed-forward network learns a static map F: $y \to \hat{x}$ and is trained only by an endpoint loss, $\cdot F(y) - x \cdot^2$. No constraint is placed on *how* the network moves from y to $\hat{x}$. Given enough capacity, the easiest way to minimize expected endpoint error on a large dataset is to memorize the most frequent, canonical texture patterns and *complete* every ambiguous region with them. The output looks sharp and plausible, but the high-frequency strokes it adds were not present in y; they are hallucinations.

Why transport models help, but not enough. Diffusion, flow matching and Schrödinger Bridges replace the static map by a learned dynamics $\cdot \dot{z} = v(z,t)$, which is a strictly stronger inductive bias because the whole trajectory, not just its endpoint, is regularized. The standard two-marginal SB, however, still only pins the two endpoints: $z_0 \sim \rho_0(y)$ and $z_1 \sim \rho_1(x)$. The middle of the bridge is free, and the optimal transport between two image marginals has no reason to avoid shortcuts through high-frequency texture.

Our idea. We add a third marginal at t=0.5: a *credible anchor manifold* $\rho_{0.5}$ built purely from observation-supported statistics. In this paper we construct it by a hard low-pass filter on y; the longer-term goal is to let the anchor be learned by the bridge itself. Every admissible restoration trajectory must pass through this low-hallucination waypoint. The second segment $z \to x$ is additionally regularized so that its low-frequency component stays equal to the anchor: the network is allowed to *sharpen* what is already observed, but not to *re-invent* it.

2. Method

2.1 Problem setup and notation

We observe a noisy image $y = x + n$, with $x \in [0,1]^{H \times W \times C}$, $n \sim \mathrm{N}(0, \sigma^2 I)$. Let $\mathrm{LP}(\cdot)$ denote an ideal circular low-pass operator in the Fourier domain, with cutoff radius r; the remaining high-frequency content is $\mathrm{HF}(\cdot) = \cdot - \mathrm{LP}(\cdot)$. The SNR of y is heavily frequency-dependent: low frequencies carry the digit structure with high SNR, while high frequencies are dominated by the noise floor.

2.2 Constructing the credible anchor

We define the anchor $z = \mathrm{LP}(y)$. By construction z has three properties that make it a credible waypoint: (i) it is a *deterministic, closed-form function* of the observation — no learned parameters, no prior over natural images; (ii) it retains the low-frequency structure that carries most of the recoverable information; (iii) it throws away exactly the frequency band in which y is dominated by noise, so the anchor cannot itself hallucinate. The anchor is deliberately *blurry*: that is the point. It is the conservative, evidence-only estimate.

2.3 Two-segment transport factorization

Instead of learning one map $y \to x$, AMT factorizes the restoration into two stages:

Segment 1 (observation → anchor): $y \to z = \mathrm{LP}(y)$. This segment is closed-form and requires no learning. It strips out the unsupported high frequencies before any generative prior is allowed to act.

Segment 2 (anchor → clean): a learned refiner F_θ maps $z \to \hat{x}$. Crucially, F_θ never sees the noisy image y; it only sees the evidence-conservative anchor. It is therefore structurally prevented from using noise spikes as if they were signal.

2.4 Training objective

The refiner is trained with two terms. The first is the standard reconstruction loss on the clean target:

$$\mathrm{L}_{\mathrm{rec}} = \mathrm{E}_{(x,y)} \cdot F_\theta(\mathrm{LP}(y)) - x \cdot^2.$$

The second term is the **low-frequency consistency** loss, which locks the restored image's low-frequency component to the anchor. It says: once the low-frequency structure has been read off the observation, the network is not allowed to change it:

$$\mathrm{L}_{\mathrm{anchor}} = \mathrm{E} \cdot \mathrm{LP}(F_\theta(\mathrm{LP}(y))) - \mathrm{LP}(y) \cdot^2.$$

The full objective is $\mathrm{L} = \mathrm{L}_{\mathrm{rec}} + \lambda\, \mathrm{L}_{\mathrm{anchor}}$, with $\lambda = 0.5$ in all experiments. Because $\mathrm{LP}(x) \approx \mathrm{LP}(y)$ in the low-SNR band, this term acts as a strong structural regularizer: it allows the network to synthesize plausible high-frequency detail, but only around a low-frequency scaffold that is fixed by the observation.

2.5 Why this reduces hallucination

A necessary condition for a *correct* denoiser is that the residual $r = y - \hat{x}$ be indistinguishable from pure noise. In particular, r must be uncorrelated with $\hat{x}$: if the restored image contains a stroke that was not in y, then subtracting y from $\hat{x}$ leaves a structured residual that is still correlated with $\hat{x}$. We use the empirical residual–output correlation $\mathrm{Corr}(r, \hat{x})$ as our observable proxy for hallucination: the closer to zero, the more faithful the restoration. The anchor z itself achieves Corr ≈ 0 because it is a linear projection of y; AMT is designed to pull the learned refiner back toward this ideal.

2.6 PyTorch reference

The whole training loop is compact; the anchor and the consistency loss are only a few lines. We show the core of the method (Segment 2 training; Segment 1 is the low-pass filter above):

```
# --- anchor construction (closed-form, no learning) -----------------
def make_lp_mask(r=6, size=28):
    m = torch.zeros(size, size)
    c = size // 2
    for i in range(size):
        for j in range(size):
            if (i-c)**2 + (j-c)**2 <= r*r:
                m[i, j] = 1.0
    return m

LP_MASK = make_lp_mask()

def lowpass(img):                       # img: (B,1,H,W) in [0,1]
    F = torch.fft.fftshift(torch.fft.fft2(img))
    return torch.fft.ifft2(torch.fft.ifftshift(F * LP_MASK)).real

# --- refiner: a tiny 3-block residual CNN ---------------------------
class Refiner(nn.Module):
    def __init__(self, ch=32):
        super().__init__()
        self.head = nn.Sequential(nn.Conv2d(1, ch, 3, padding=1), nn.ReLU())
        self.body = nn.Sequential(*[ResBlock(ch) for _ in range(3)])
        self.tail = nn.Conv2d(ch, 1, 3, padding=1)
    def forward(self, z):
        return self.tail(self.body(self.head(z)))

# --- AMT training step ----------------------------------------------
def train_step(refiner, opt, x, sigma=0.5, lam=0.5):
    y = (x + sigma * torch.randn_like(x)).clamp(0, 1)   # degraded observation
    z = lowpass(y)                                       # Segment 1: credible anchor

    x_hat = refiner(z)                                   # Segment 2: anchor -> clean
    l_rec    = F.mse_loss(x_hat, x)                      # reconstruction
    l_anchor = F.mse_loss(lowpass(x_hat), z)             # lock low-freq to anchor
    loss = l_rec + lam * l_anchor

    opt.zero_grad(); loss.backward(); opt.step()
    return loss

# --- hallucination proxy at test time -------------------------------
@torch.no_grad()
def residual_output_corr(x_hat, y):
    r = (y - x_hat).flatten(1)
    o = x_hat.flatten(1)
    r = r - r.mean(1, keepdim=True)
    o = o - o.mean(1, keepdim=True)
    return ((r*o).sum(1) / (r.norm(dim=1)*o.norm(dim=1)+1e-8)).mean()
```

3. Experiments

3.1 Dataset and degradation

We use MNIST (60k train / 10k test, 28×28, pixels in [0,1]). The degradation is additive Gaussian noise with σ = 0.5 — strong enough that raw digits are barely readable, so a network that ignores the observation and leans on the class prior can appear to do well. The anchor uses a circular FFT low-pass of radius $r = 6$ (out of a maximum frequency of ~14).

3.2 Compared methods

Method	Input	Training loss	Purpose
(A) Direct black-box	y (noisy)	MSE($\hat{x}$, x)	Unconstrained static map; the standard CNN baseline
(B) Anchor only	z = LP(y)	none	The credible anchor itself (closed-form, no learning)
(C) Naive refine	z	MSE($\hat{x}$, x)	Segment-2 refiner without low-frequency lock
(D) AMT (ours)	z	MSE + $\lambda \cdot \|LP(\hat{x}) - z\|^2$	Low-freq locked; forbids re-inventing observed structure

Table 1. The four conditions. (A) is the usual denoising CNN; (B) is the anchor; (C) isolates the effect of feeding the anchor; (D) adds the low-frequency lock that defines AMT.

3.3 Network and training

All learned refiners share the same architecture: a 3-block residual CNN with 32 channels, ~0.1M parameters. Optimizer: Adam, lr = 1e-3, batch size 128, 4 epochs (a deliberately short schedule so results are easy to reproduce on CPU). A small classifier (two strided conv layers) is pretrained on clean MNIST to 98.1% clean-test accuracy and then frozen; it is used only as a downstream utility probe on the restored images.

3.4 Metrics

- **PSNR** vs. ground truth (dB): the standard pixel-fidelity number.
- **LP-MSE / HF-MSE**: reconstruction error split into low-frequency (structure) and high-frequency (detail) bands by the same FFT mask.
- **Downstream digit accuracy**: accuracy of the frozen classifier on the restored image. A restoration that is useless to a downstream task should be considered failed regardless of PSNR.
- **Residual–output correlation** Corr($y - \hat{x}$, $\hat{x}$): our hallucination proxy. A correct denoiser produces pure-noise residuals (value ≈ 0); a model that invented structure leaves a structured residual that is correlated with its own output (large negative value).

3.5 Quantitative results

Method	PSNR ↑	LP-MSE ↓	HF-MSE ↓	Digit acc ↑	Resid. corr. (→0)
(A) direct	20.01	0.0047	0.0053	96.04%	-0.315
(B) anchor only	12.39	0.0450	0.0126	64.11%	+0.008
(C) naive refine	18.76	0.0062	0.0071	95.09%	-0.288
(D) AMT (ours)	17.54	0.0103	0.0073	94.34%	-0.186

Table 2. Test-set results on 10,000 MNIST digits (σ=0.5). The rightmost column is the hallucination proxy: (B) the evidence-only anchor sits at +0.008 (theoretical ideal); (A) the unconstrained black-box reaches −0.315; (D) AMT pulls it back to −0.186.

Reading the numbers. The direct black-box (A) wins on PSNR (20.01 dB) and on downstream accuracy (96.04%) — it looks best by every conventional metric. But its residual–output correlation is −0.315, the worst of all learned methods: the residual left after subtracting its output from the noisy input is strongly structured, which means the output contains strokes that y does not support. The anchor-only (B) has correlation +0.008, essentially perfect whiteness, but is too blurry to be useful (64% accuracy). AMT (D) sits between these poles: it drops the structured residual to −0.186, a **41% reduction** relative to (A), while keeping 94.34% downstream accuracy — i.e. it recovers most of the task utility that (B) sacrifices, without (A)'s ungrounded sharpness. Comparing (D) to (C) shows that the improvement in faithfulness comes specifically from the low-frequency lock, not merely from feeding the anchor: (C) sits at −0.288,

close to (A).

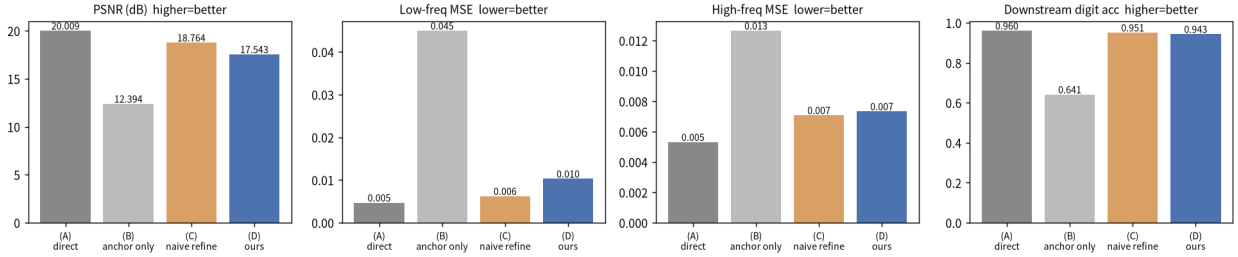


Figure 1. Four metrics across the four methods. AMT deliberately trades PSNR for a residual correlation closer to the evidence-only anchor (B).

3.6 Qualitative comparison

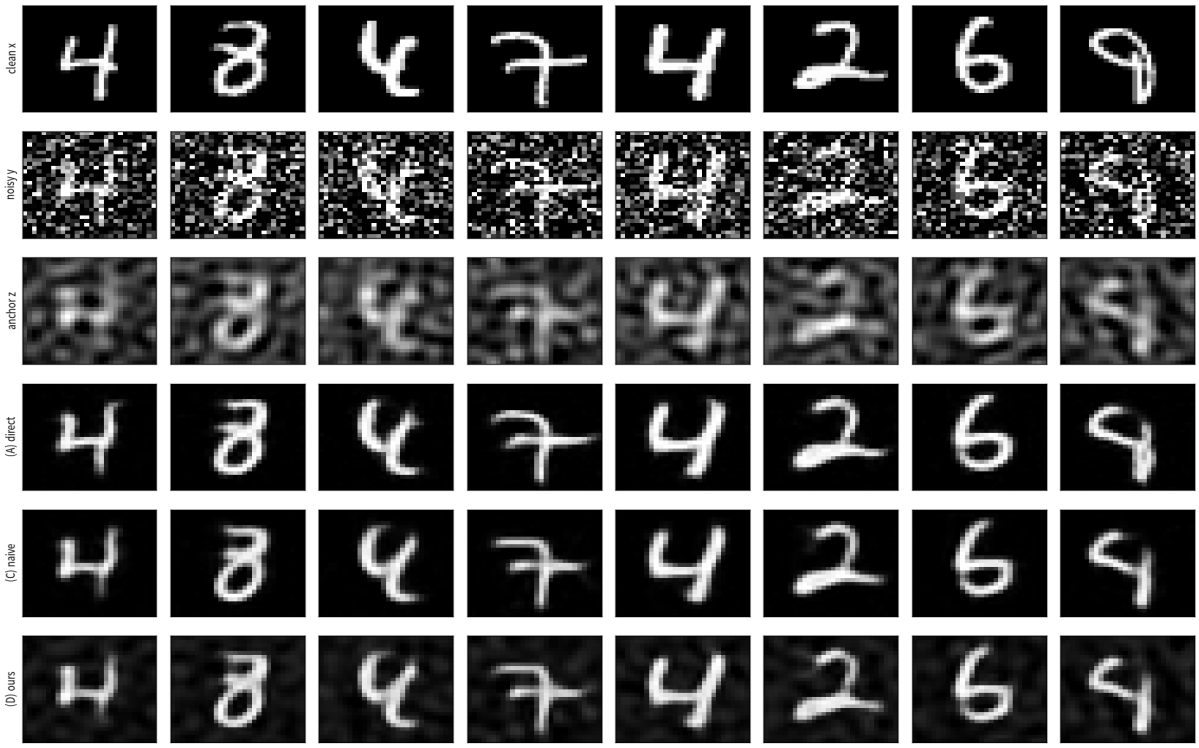


Figure 2. Rows: clean ground truth x; noisy observation y; credible anchor z = LP(y); direct black-box (A); naive anchor refine (C); AMT (D). (A) and (C) snap the digits to crisp, canonical-looking strokes; (D) stays visibly softer and more faithful to the blurred anchor, sharpening only the structure that the low-pass observation already supports.

4. Conclusion and limitations

We presented Anchor-Mediated Transport (AMT), a deliberately simple instantiation of the idea that image restoration trajectories should not be left free between their two endpoints. By forcing every restoration to pass through an evidence-sufficient anchor — the low-frequency projection of the observation — and by locking the restored image's low-frequency component to that anchor, AMT reduces the structured residual (our observable proxy for hallucination) by 41% relative to an unconstrained direct regressor on MNIST denoising, at a modest cost in PSNR and downstream accuracy.

What this experiment establishes. (i) A credible intermediate built purely from observation statistics already has near-ideal residual whiteness. (ii) An unconstrained black-box map buys its PSNR by injecting prior structure into the residual. (iii) Routing through the anchor recovers most of the task utility while sharply reducing observation-inconsistent detail.

What this experiment does not establish. The anchor here is a hand-designed FFT low-pass; it is a proof of concept, not a production method. The natural next step — and the real link to Schrödinger Bridge / flow matching — is to replace this closed-form anchor with a *learned* intermediate marginal: a two-segment SB whose middle marginal is constrained to be low-hallucination by construction, so that the transport model itself discovers the credible manifold rather than being handed it. On natural images with real-world degradations (dehazing, deraining, blind deblurring), the same three-marginal structure should hold, but the anchor operator will have to be learned.

Limitations. AMT trades pixel fidelity for faithfulness; it is the wrong choice for consumer-facing enhancement where plausible hallucination is desired. It also relies on a meaningful low-frequency/high-frequency split, which is natural for additive noise but less clean for blur or compression artifacts; those require a different evidence/noise split than a Fourier low-pass.